Assignment 1: Square Numbers – Iteration

Assignment for COSC 1100 – Intro to Programming

Prior to attempting this problem, you should have done the following:

1. View the Content, including assigned readings for weeks 1 through 6.
2. Viewed the Control Structures - Iteration videos from week 6 and completed the associated Pre-Class Activity.

General Requirements

1. This assignment is to be completed **individually**. Group submissions will not be considered.
2. Analyze the problem, design a documented plan (i.e. flowchart or pseudo-code), code and test a solution **following the step-by-step approach** presented in this course.
3. Submit your solution and plan to the appropriate assignment folder on DC Connect **by the due date provided**.
4. Your instructor will assign a grade and post feedback on your submission to DC Connect.

Program Requirements

For this lab you will create a simple console application using Python that will graphically display all the square numbers within a range provided by the user. For our purposes, a square number is any number that can be computed by multiplying a whole number by itself. The first several square numbers are 1 (1 squared), 4 (2 squared), 9 (3 squared) and 16 (4 squared). This assignment will provide you with the opportunity to design program logic with both individual and nested loops. It should also reinforce the time-saving value of planning before you code.

Getting User Input:

In the input phase of the program, display a banner-like header and then prompt the user for the bottom limit and the top limit to the numbers they want to investigate. The lower limit must be a whole number between the range of 1 and 10000 inclusive, and the upper limit must be between the lower limit and 10000 inclusive. Should the user enter something other than a valid, in-range number, then display an appropriate message that describes the problem and give the user the opportunity to input the limit again.

You will need to use “try … except” or another means verifying whether the user input is numeric and a whole number. Also, make sure that no processing occurs until the application has valid input.

The example below shows multiple invalid inputs; match this as closely as possible:

```
**********************************************************************
                        SQUARE NUMBER FINDER
**********************************************************************

Please enter the lower bound: Chicken
Invalid input: Please enter a whole number.

Please enter the lower bound: 4.5
Invalid input: Please enter a whole number.

Please enter the lower bound: 10001
Invalid input: Please enter a number between 1 and 10000.

Please enter the lower bound:
```

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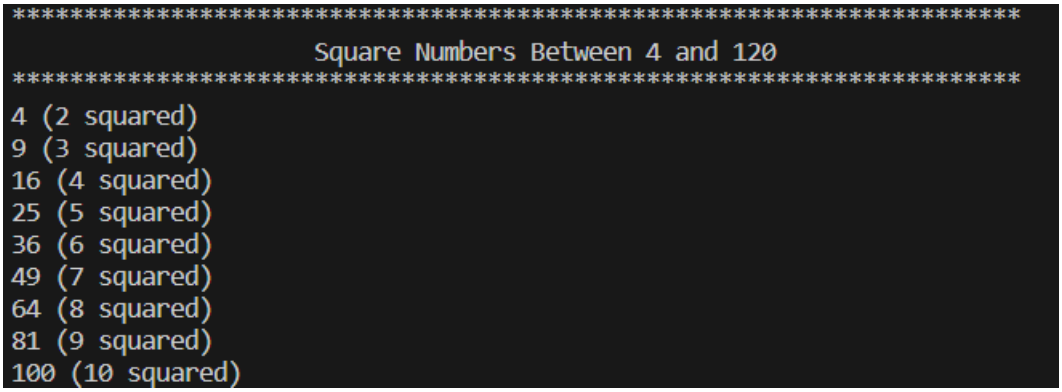
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Final Output:

Once you have valid input, you will produce a list of all square numbers between the user's entered lower bound and upper bound. Start by displaying "Square numbers between", including the boundaries entered. In addition to displaying the square number, you will also display the square root of the number that it is the square of.

The following output example shows what your program should produce if the user entered boundaries were 4 and 120; match this as closely as possible:



At the end, be sure to use `input()` or something similar to wait before closing the program.

Style Guide Notes

To be eligible for full marks on this or any assignment in this course your application must conform to the requirements as outlined above as well as our prescribed style guide, in this case making sure to observe the PEP8 naming conventions for Python as well as appropriate and complete program documentation.

Development Hints

You may be tempted to utilize parts of your existing work or demo code as a starting point for this project. Expect differences! It is strongly recommended that you code this assignment following the "from plan to program" process from the beginning.

Remember that the aim of this lab is to have you practice using loops. There is more than one opportunity to do so in this problem. An efficient, working solution will probably involve at least three different loops. If you find yourself writing 10+ consecutive if/elif blocks you are taking the wrong approach.

Academic Integrity

By submitting this assignment to DC Connect for evaluation, you agree to the following terms:

- ☐ This work is your original work, with the majority its contents originally created by yourself.
- ☐ Any aspects of this work that were created by or based heavily on reference to another person's work, a website, a video, or a program (including generative AI systems) are cited with suitable identification to observe the original source.
- ☐ You did not transmit this file to anyone else's computer, nor receive any part of this work from anyone else's computer.

Failure to observe these points will leave you subject to Durham College's Academic Integrity Policy, with a minimum penalty of a grade of 0 on the assignment and an academic integrity alert submitted to the faculty office.

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